

Homework #5

Chapter 8

22. A periodic signal $x(t)$ with a fundamental period of 2 seconds is described over one fundamental period by

$$x(t) = \begin{cases} \sin(2\pi t), & |t| < 1/2 \\ 0, & 1/2 < |t| < 1 \end{cases}$$

Graph the signal and find its complex CTFS description. Then graph on the same scale approximations to the signal $x_N(t)$ given by

$$x_N(t) = \sum_{k=-N}^N X[k] e^{j2\pi(kf_F)t}$$

- for $N = 1, 2$ and 3 . (In each case, the time scale of the graph should cover at least two fundamental periods of the original signal.)
23. Using MATLAB, graph the following signals over the time range $-3 < t < 3$.

- (a) $x_0(t) = 1$
- (b) $x_1(t) = x_0(t) + 2\cos(2\pi t)$
- (c) $x_2(t) = x_1(t) + 2\cos(4\pi t)$
- (d) $x_{20}(t) = x_{19}(t) + 2\cos(40\pi t)$

For each part, (a) through (d), numerically evaluate the area of the signal over the time range $-1/2 < t < 1/2$.

30. Each signal in figure E.30 is graphed over a range of exactly one fundamental period. Which of the signals have harmonic functions $X[k]$ that have a purely real value for every value of k ? Which have a purely imaginary value for every value of k ?

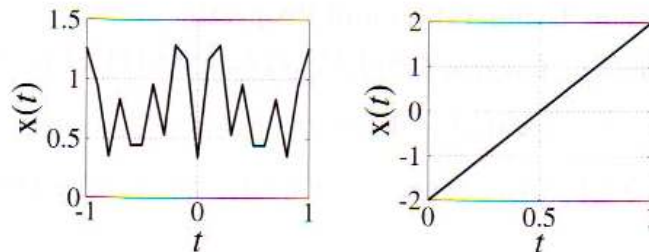


Figure E.30

32. In some types of communication systems, binary data are transmitted using a technique called binary phase-shift keying (BPSK) in which a 1 is represented by a burst of a sine wave and a 0 is represented by a burst that is the exact negative of the burst that represents a 1. Let the sine frequency be 1 MHz and let the burst width be 10 periods of the sine wave. Find and graph the CTFS harmonic function for a periodic binary signal consisting of alternating 1's and 0's, using its fundamental period as the representation time.

Chapter 12

32. Draw asymptotic and exact magnitude and phase Bode diagrams for the frequency responses of the circuits and systems in figure E.32.

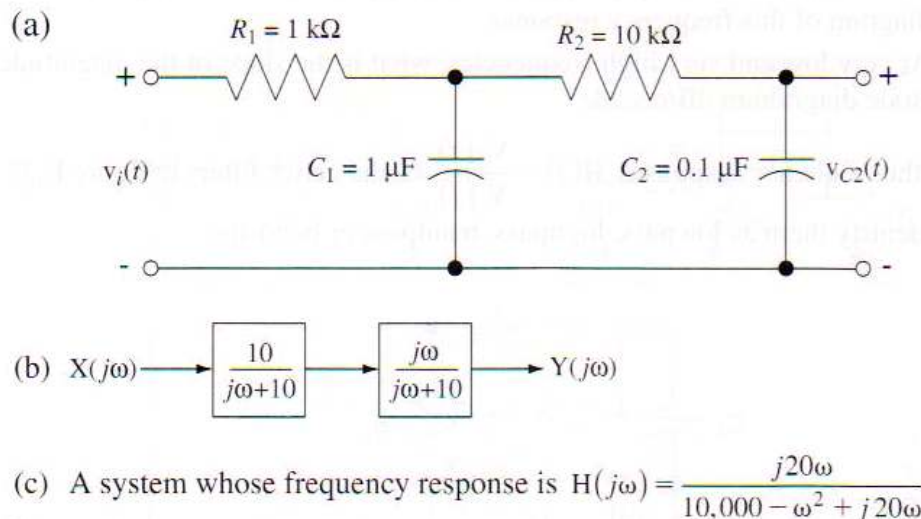


Figure E.32